INCH-POUND

MIL-M-28787/293(SH)
AMENDMENT 2
26 November 1991
SUPERSEDING
AMENDMENT 1
6 July 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE CHH

This amendment forms a part of Military Specification MIL-M-28787/293(EC), dated 7 March 1980, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2, second line: Delete "100°C" and substitute "85°C".

PAGES 4 and 5

4.3: Delete and substitute the following:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

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PAGE 5

* 4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. Test conditions A, B, C, D, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, and F where applicable, followed by the remaining tests in the sequence."

* 4.5.1, fifth and sixth lines: Delete "test conditions D through X" and substitute "test conditions G through X".

* 4.5.3, third line: Delete "D through X" and substitute "E and G through X".

* 4.5.3, sixth line: Delete "D through Y" and substitute "E and G through X".

PAGE 7

* TABLE II, Test conditions D1-D8, and F1-F28, Test type column: Delete "X" and substitute "S".

PAGE 8

* Delete and substitute new page as printed on page 4 of this amendment.

PAGES 7, 8, and 9

TABLE II, Class II column heading: Delete "100°" (three places) and substitute "85°" (three places).

PAGE 17

FIGURE 6: Delete and substitute new figure 6 as printed on page 5 of this amendment.

PAGE 29

FIGURE 10: Delete and substitute new figure 10 as printed on page 6 of this amendment.

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PAGE 33

FIGURE 11: Delete and substitute new figure 11 as printed on page 7 of this amendment.

PAGE 56

FIGURE 17: Delete and substitute new figure 17 as printed on page 8 of this amendment.

PAGE 69

FIGURE 23: Delete and substitute new figure 23 as printed on page 9 of this amendment.

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy – SH

(Project 5963–N758)

Agent: NW

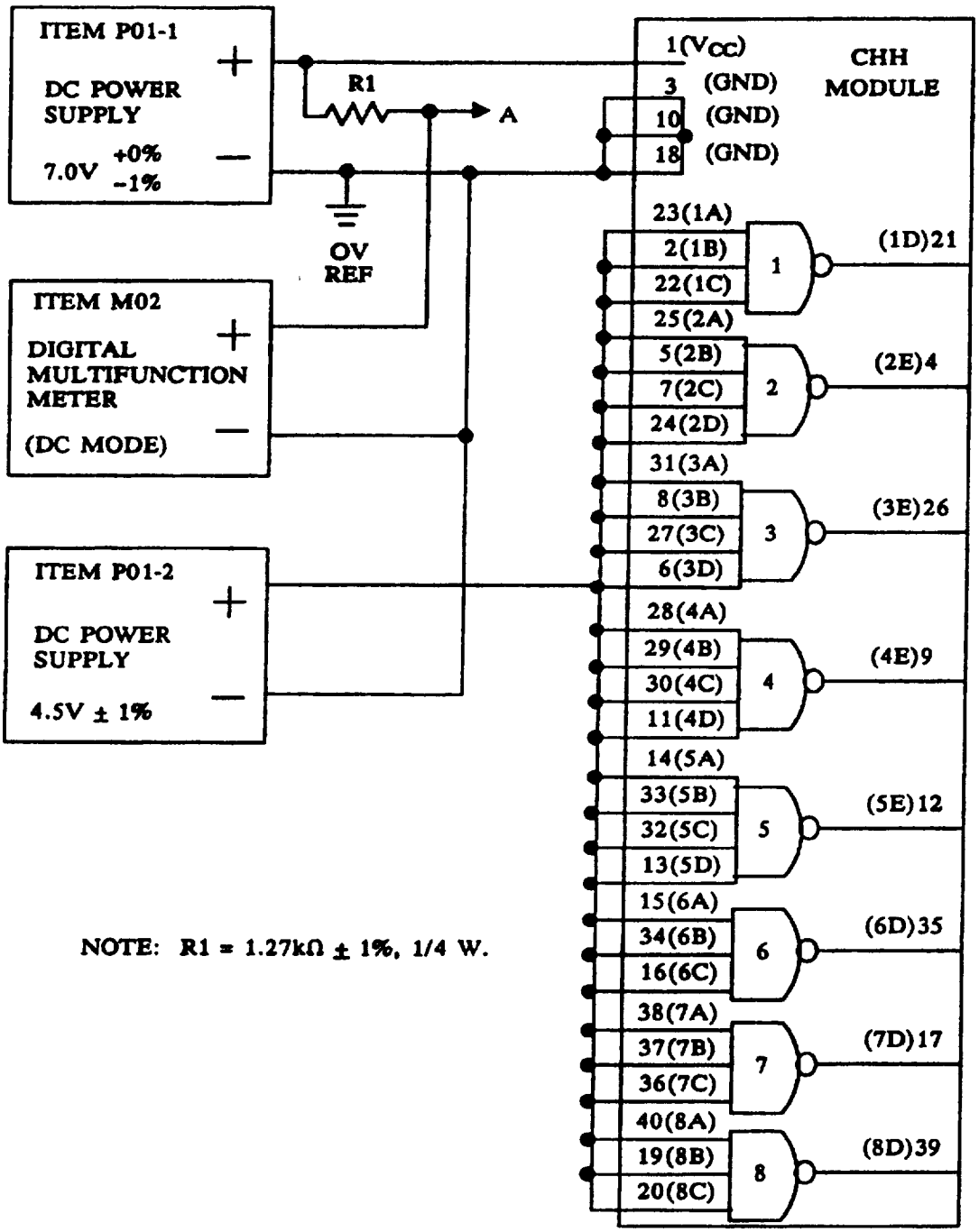
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TABLE II. Module electrical requirements. -Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5° 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N9-N16	Transition time (t_{TLH}) C_L = 150 pF	XV	Q	10.0	80.0	10.0	80.0	10.0	90.0	10.0	80.0	10.0	100.0	ns
P1-P8	Transition time (t_{THL}) C_L = 50 pF	XV	S	3.0	18.0	3.0	18.0	3.0	20.0	3.0	18.0	3.0	22.0	ns
P9-P16	Transition time (t_{THL}) C_L = 150 pF	XV	Q	6.0	40.0	6.0	40.0	6.0	42.0	6.0	40.0	6.0	44.0	ns
Q1-Q28	Propagation delay time (t_{PLH}) C_L = 50 pF	XV	S	3.0	23 0	3.0	23.0	3 0	25.0	3.0	27.0	3.0	27.0	ns
Q29-Q56	Propagation delay time (t_{PLH}) C_L = 150 pF	XV	Q	4.0	33.0	4.0	33.0	4.0	35.0	4.0	37.0	4.0	37.0	ns
R1-R28	Propagation delay time (t_{PHL}) C_L = 50 pF	XV	S	3.0	23.0	3.0	23.0	3.0	25.0	3.0	27.0	3.0	27.0	ns
R29-R56	Propagation delay time (t_{PHL}) C_L = 150 pF	XV	Q	4.0	33.0	4.0	33.0	4.0	35.0	4.0	37.0	4.0	37.0	ns
S1-S4	Output to output delta propagation delay time (Δt_{PLH}) C_L = 50 pF	XVI	S	-3.0	3.0	-3.0	3.0	-3.0	3.0	-3.0	3.0	-3.0	3.0	ns
T1-T4	Output to output delta propagation delay time (Δt_{PHL}) C_L = 50 pF	XVI	S	-3.0	3.0	-3.0	3.0	-3.0	3.0	-3.0	3.0	-3.0	3.0	ns
U1-U8	Output crosstalk, low level	XVII	Q	0	500	0	500	0	500	0	500	0	500	mV
V1-V8	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

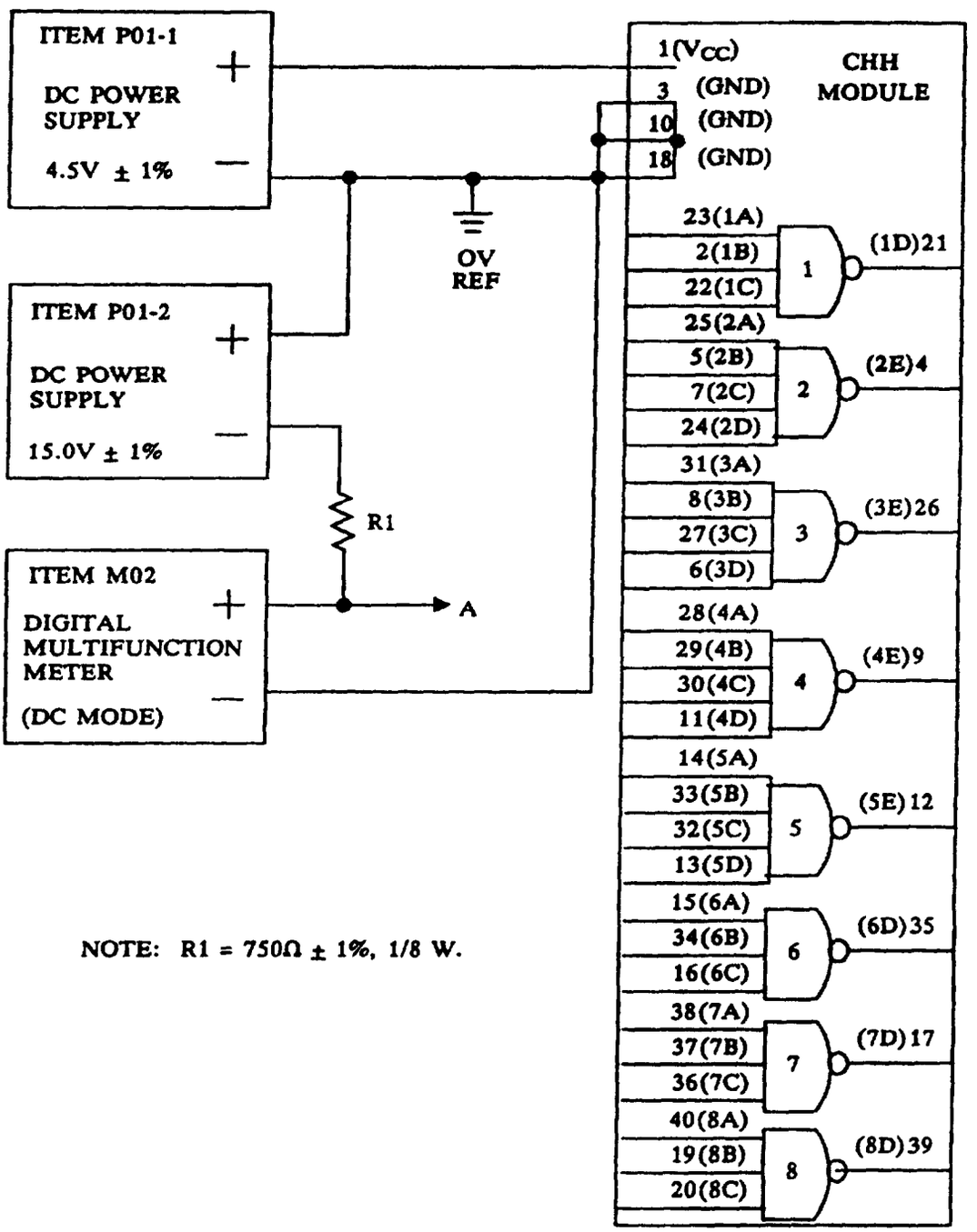
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NOTE: R1 = 1.27kΩ ± 1%, 1/4 W.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

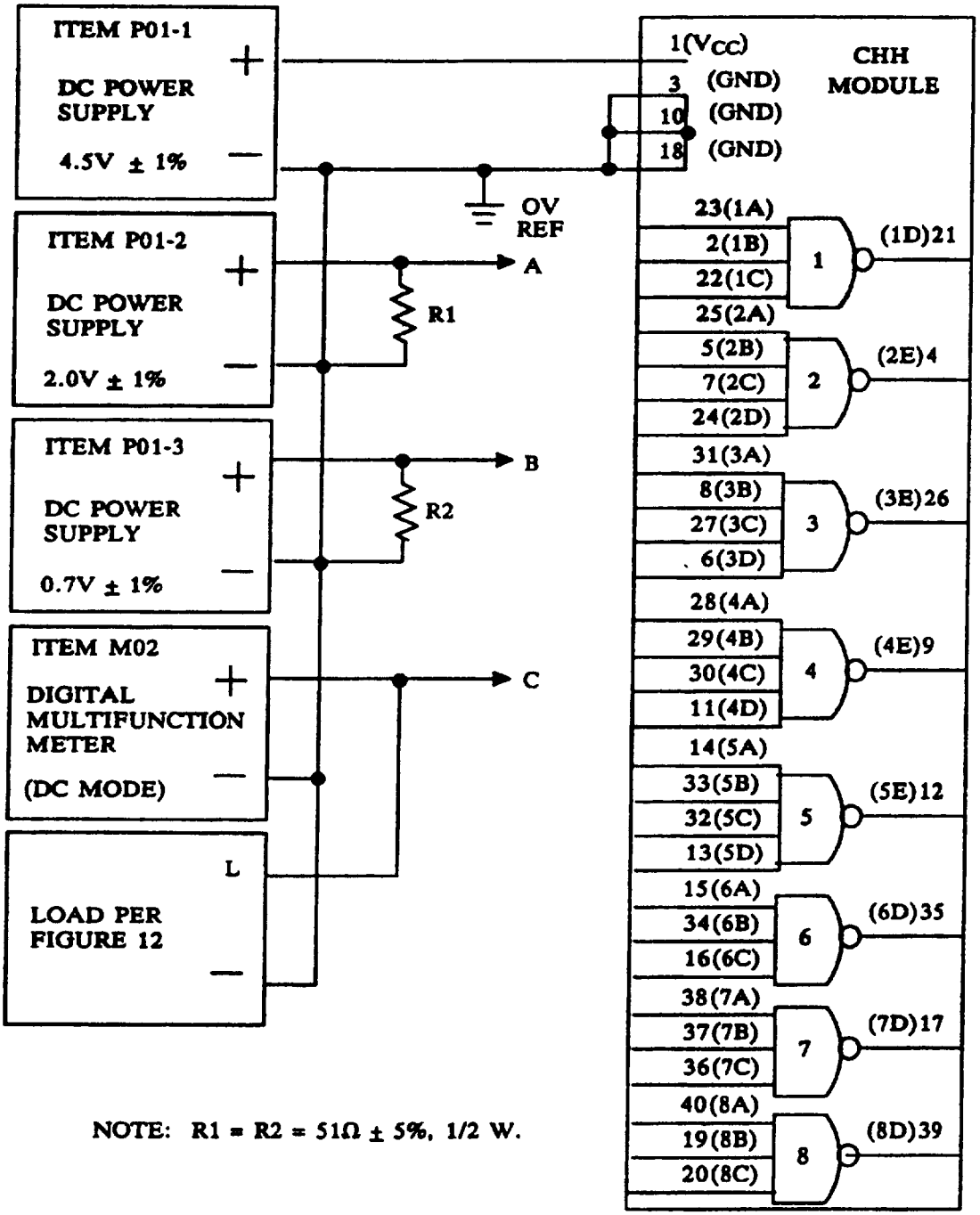
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NOTE: R1 = 750Ω ± 1%, 1/8 W.

FIGURE 10. Input clamp diode voltage test circuit for table VIII.

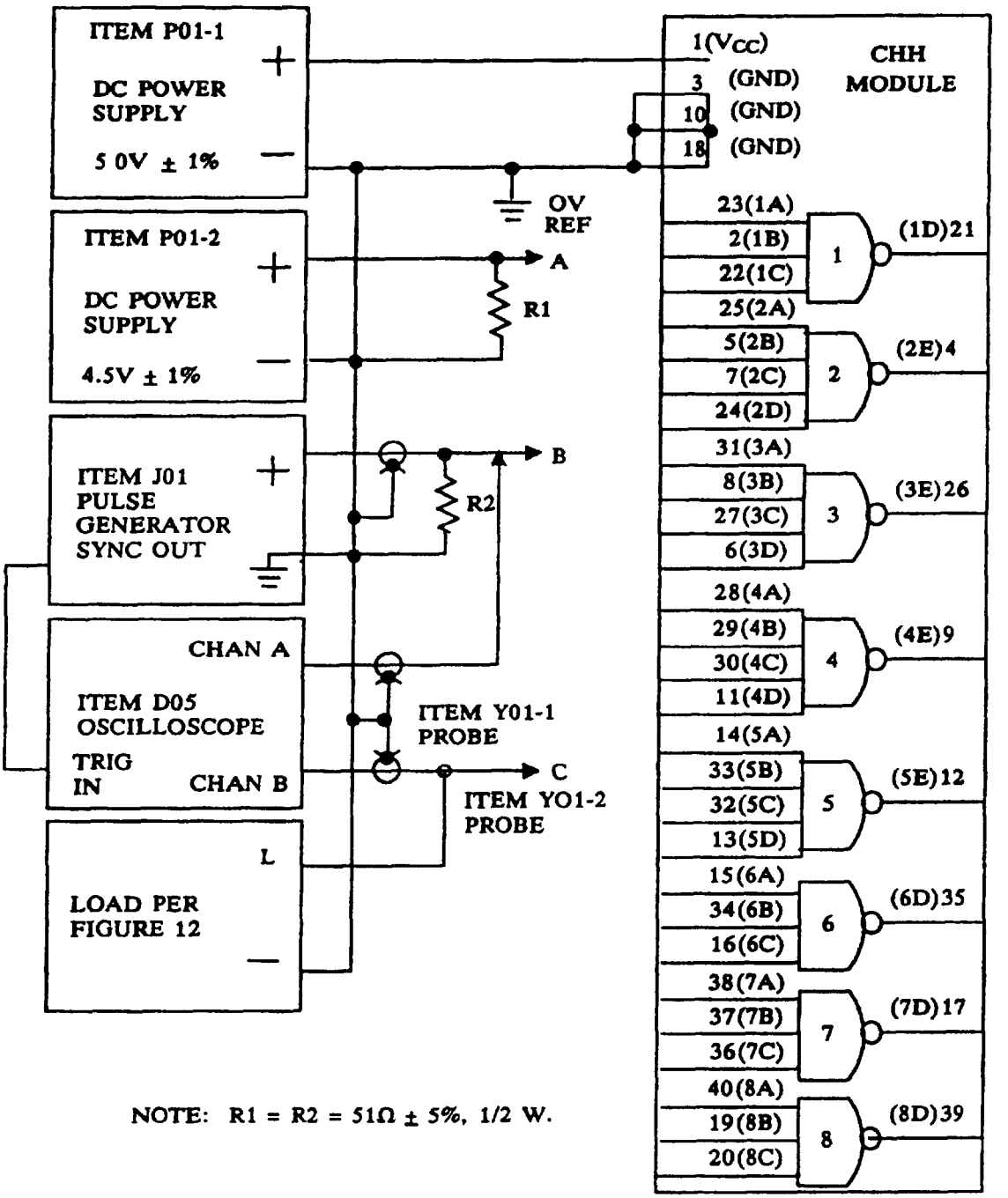
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NOTE: R1 = R2 = 51Ω ± 5%, 1/2 W.

FIGURE 11. Static output voltage (low-level and high-level) test circuit for tables IX and X.

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NOTE: R1 = R2 = 51Ω ± 5%, 1/2 W.

FIGURE 17. Transition time and propagation delay tume test circuit for table XV.

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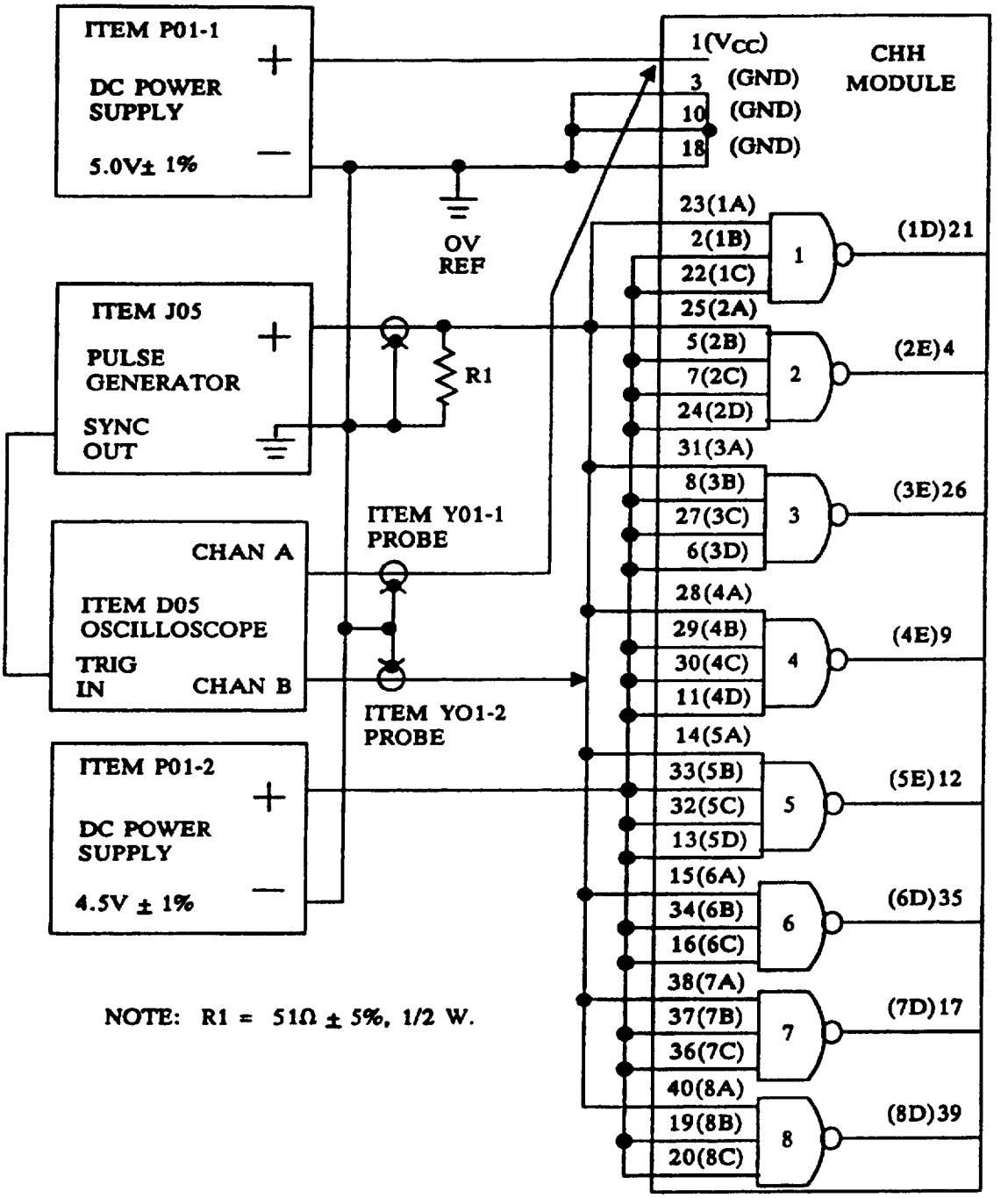


FIGURE 23. Power supply transient amplitude test circuit for table XVIII.

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